particle and as seen by external observers is completely analogous to the difference between a stellar implosion as seen on the star's surface (rapid plunge through the horizon) and as seen by external observers (freezing of the implosion; last part of Chapter 6).

The second particle (Figure 7.Bb) is thrown toward the hole on an inspiraling trajectory that rotates oppositely to the hole's spin. However, as the particle spirals closer and closer to the horizon, the swirl of space grabs hold of it and reverses its rotational motion. Like the first particle, it is forced into lockstep rotation with the horizon, as seen by external observers.

Besides creating a swirl in space, the spin of a black hole also distorts the hole's horizon, in much the same way as the spin of the Earth distorts the Earth's surface. Centrifugal forces push the spinning Earth's equator outward a distance of 22 kilometers relative to its poles. Similarly, centrifugal forces make a black hole's horizon bulge out at its equator in the manner depicted in Figure 7.9. If the hole does not spin, its horizon is spherical (left half of figure). If the hole spins rapidly, its horizon bulges out strongly (right half of figure).

If the hole were to spin extremely rapidly, centrifugal forces would tear its horizon apart much like they fling water out of a bucket when the bucket spins extremely rapidly. Thus, there is some maximum spin rate at which the hole can survive. The hole on the right half of Figure 7.9 is spinning at 58 percent of this maximum.

Is it possible to spin a hole up beyond its maximum allowed rate, and thereby destroy the horizon and catch a glimpse of what is inside? Unfortunately not. In 1986, a decade after the golden age, Werner Israel showed that, if one tries to make the hole spin faster than its maximum by any method at all, one will always fail. For example, if one tries to speed up a maximally spinning hole by throwing fast-spinning matter into it, centrifugal forces will prevent the fast-spinning matter from reaching the horizon and entering the hole. More to the point, perhaps, any tiny random interaction of a maximally spinning hole with the surrounding Universe (for example, the gravitational pull of distant stars) acts to slow the spin a bit. The laws of physics, it seems, don't want to let anyone outside the hole peek into its interior and discover the quantum gravity secrets locked up in the hole's central singularity (Chapter 13).